

Name: Answer Key Group: _____

Midterm exam (1h15)

Exercise 1 (4 points) Consider the following sample, made up of draws according to a normal distribution with expectation μ and standard deviation $\sigma=3$.

14.5	9.3	12.3	10.4	12.9	10.2	14.2	13.5
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1. Construct a confidence interval for μ , at the 99% confidence level.

We have $n=8$ and $\bar{X} = \frac{1}{8} \sum_{i=1}^8 x_i = \frac{97.3}{8} \Rightarrow \bar{X} = 12.16$ (9.5)

Since σ is known, we use the statistics $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$ (9.5)

and for $\alpha=0.01$, the confidence interval is

$$I_{0.99} = \left[\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}; \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right] \quad ; \quad z_{1-\frac{\alpha}{2}} = z_{0.995} = 2.575 \quad (9.5)$$

$$\Rightarrow I_{0.99} = \left[12.16 - \frac{2.575 \cdot 3}{\sqrt{8}}; 12.16 + \frac{2.575 \cdot 3}{\sqrt{8}} \right]$$

$$\Rightarrow I_{0.99} = [9.43; 14.89] \quad (1)$$

2. Still at the 99% confidence level, we want to obtain a margin of error for μ of less than 0.5; how do we choose the sample size?

$$\text{We have } e \leq 0.5 \quad \text{and} \quad e = z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \Leftrightarrow \frac{\sqrt{n}}{3\sigma} \geq \frac{1}{e} \quad (9.5)$$

$$\Leftrightarrow n \geq \left(\frac{3\sigma}{e} \right)^2 \Leftrightarrow n \geq \left(\frac{2.575 \cdot 3}{0.5} \right)^2 \Leftrightarrow n \geq 237.77$$

$$\text{Then } \boxed{n \geq 238} \quad \text{for } e \leq 0.5 \quad (9.5)$$

Exercise 2 (6 points)

In a university, there is a proportion p of boys and a proportion $1-p$ of girls. We draw a sample of $n=100$ students (with replacement) enrolled at this university, and note $X_1, X_2, \dots, X_n$ the result of each draw. For each student drawn, the random variable X is equal to 1 if the student drawn is a boy and 0 if it is a girl.

1. Give the distribution of X .

We have $X_i = \begin{cases} 1 & \text{if it's a boy} \\ 0 & \text{if it's a girl} \end{cases}$

and $P(X_i=1)=p$ and $P(X_i=0)=1-p$

then $X_i \sim B(p)$ $\Rightarrow E[X_i]=p$

$$Var(X_i)=p(1-p)$$

2. Let $Y = \sum_{i=1}^n X_i$. What is the distribution of Y , determine its expectation and variance.

$\{X_i\}_{i=1}^n$ is an i.i.d sample of distribution $B(p)$

$$\text{or } Y = \sum_{i=1}^n X_i \Rightarrow Y \sim B(n, p) \quad (95)$$

$$\text{Then } E[Y] = np \quad (925)$$

$$Var(Y) = np(1-p) \quad (925)$$

3. Let $F_n = Y/n$, be an unbiased estimator of the proportion p . As a function of F_n give a confidence interval for p at the 95% confidence level (Numerical application: $Y = 60$).

$$F_n = \frac{Y}{n} = \bar{X} \Rightarrow E[F_n] = E[\bar{X}] = E[X] = p \quad (95)$$

$$\text{and } Var(F_n) = Var(\bar{X}) = \frac{Var(X)}{n} = \frac{p(1-p)}{n} \quad (95)$$

$$\text{Since } n > 30 \quad \text{then } U_n = \frac{F_n - E[F_n]}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1) \quad (95)$$

Thus the confidence interval for p is

$$I_{\alpha} = \left[F_n - z_{1-\frac{\alpha}{2}} \sqrt{\frac{F_n(1-F_n)}{n}} ; F_n + z_{1-\frac{\alpha}{2}} \sqrt{\frac{F_n(1-F_n)}{n}} \right] \quad (95)$$

$$F_n = \frac{60}{100} = 0.6 \Rightarrow I_{0.05} = \left[0.6 - 1.96 \sqrt{\frac{0.6 \cdot 0.4}{100}} ; 0.6 + 1.96 \sqrt{\frac{0.6 \cdot 0.4}{100}} \right]$$

$$\Rightarrow I_{0.05} = [0.504 ; 0.696] \quad (1)$$

4. What must be the sample size n to know p with an absolute error of less than 1% with a confidence level equal to 0.95?

$$|e| < 0,01 \Leftrightarrow z_{1-\frac{\alpha}{2}} \sqrt{\frac{F_n(1-F_n)}{n}} < 0,01$$

$$\Leftrightarrow n > \frac{F_n(1-F_n) \cdot z_{1-\frac{\alpha}{2}}^2}{0,01^2} \quad (95)$$

$$\Leftrightarrow n > \frac{0,4 \cdot 0,4 \cdot 1,96^2}{0,01^2}$$

$$\Leftrightarrow n > 9219 \quad \Rightarrow \quad \boxed{n \approx 9220} \quad (95)$$

Exercise 3 (10 points)

Let X be a random variable with the density function:

$$f(x) = \begin{cases} \theta x^2 & \text{if } x \in [-1, 0] \\ 1 - \theta x^2 & \text{if } x \in [0, 1] \\ 0 & \text{elsewhere} \end{cases}$$

where θ is a parameter between 0 and 1.

- 1) Calculate $E[X]$ and $\text{Var}[X]$.

$$E[X] = \int_{\mathbb{R}} x f(x) dx = \int_{-1}^0 \theta x^3 dx + \int_0^1 (1 - \theta x^2) x dx$$

$$= \left. \frac{\theta x^4}{4} \right|_{-1}^0 + \left. \frac{x^2}{2} - \frac{\theta x^4}{4} \right|_0^1 = -\frac{\theta}{4} + \frac{1}{2} - \frac{\theta}{4} = \frac{1}{2} - \frac{\theta}{2}$$

$$\Rightarrow \boxed{E[X] = \frac{1}{2}(1 - \theta)} \quad (1)$$

$$E[X^2] = \int_{-1}^0 \theta x^4 dx + \int_0^1 x^2 (1 - \theta x^2) dx = \left. \frac{\theta x^5}{5} \right|_{-1}^0 + \left. \frac{x^3}{3} - \frac{\theta x^5}{5} \right|_0^1$$

$$= \frac{\theta}{5} + \frac{1}{3} - \frac{\theta}{5} \Rightarrow E[X^2] = \frac{1}{3} \quad (95)$$

Then $\text{Var}(X) = E[X^2] - E[X]^2$.

$$= \frac{1}{3} - \frac{1}{4}(1 - \theta)^2 = \frac{4 - 3(1 - 2\theta + \theta^2)}{12}$$

$$\Rightarrow \boxed{\text{Var}(X) = \frac{1 + 6\theta - 3\theta^2}{12}} \quad (95)$$

2) We consider a sample $X_1, X_2, \dots, X_n$ from the random variable X .

a) By estimating the mean of variable X by the sample mean, show that $T = 1 - 2\bar{X}$ is an estimator of the parameter θ .

$\bar{X}$ is an estimator of $E[X] = \frac{1-\theta}{2}$

$\Rightarrow 2\bar{X}$ is an estimator of $1-\theta$.

and the $1 - 2\bar{X}$ is an estimator of θ . (1)

So $T = 1 - 2\bar{X}$ is an estimator of θ .

b) Is this estimator unbiased? Convergent?

$$E[T] = E[1 - 2\bar{X}] = 1 - 2E[\bar{X}]$$

$$= 1 - 2E[X] + 1 = 1 - 2\left(\frac{1-\theta}{2}\right) + 1$$

$\Rightarrow E[T] = \theta$ T is unbiased

(0.5)

$$\lim_{n \rightarrow \infty} V(T) = \lim_{n \rightarrow \infty} V(1 - 2\bar{X}) = \lim_{n \rightarrow \infty} 4 \text{Var}(\bar{X})$$

$$= \lim_{n \rightarrow \infty} 4 \cdot \frac{\text{Var}(X)}{n} = \lim_{n \rightarrow \infty} \frac{4}{n} \left(\frac{1 + 6\theta - 3\theta^2}{12} \right) = 0$$

(0.5)

Thm. T converges to θ .

3) K is the number of random variables X_i in the sample taking only a value between -1 and 0 .

a) Show that K follows a binomial distribution whose parameters are to be determined.

$$\text{Let } Y_i = \begin{cases} 1 & \text{if } X_i \in [-1, 0] \\ 0 & \text{elsewhere} \end{cases}$$

$$P(Y_i = 1) = P(-1 \leq X_i < 0) = \int_{-1}^0 f(x) dx = \int_{-1}^0 \theta x^2 dx = \left. \frac{\theta x^3}{3} \right|_{-1}^0 = \frac{\theta}{3}$$

$$\text{and } P(Y_i = 0) = 1 - \frac{\theta}{3}$$

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$\Rightarrow$ K is the number of realization of $\{Y_i = 1\}$ during n repetitions of i.i.d. Bernoulli trials. Thus

$$K \sim B(n, \frac{\theta}{3})$$

(95)

b) Show that the random variable $W = \frac{3K}{n}$ is an unbiased estimator of θ .

We have $E[K] = \frac{n\theta}{3}$

and $E[W] = E\left[\frac{3K}{n}\right] = \frac{3}{n} \cdot \frac{n\theta}{3}$

(45)

$\Rightarrow E[W] = \theta$

Thus W is an unbiased estimator of θ .

c) Is W convergent?

$$\text{Var}(K) = n \cdot \frac{\theta}{3} \left(1 - \frac{\theta}{3}\right) = \frac{n\theta(3-\theta)}{9}$$

Then $\lim_{n \rightarrow \infty} \text{Var}(W) = \lim_{n \rightarrow \infty} \text{Var}\left(\frac{3}{n} K\right) = \lim_{n \rightarrow \infty} \frac{9}{n^2} \text{Var}(K)$

(1)

$$= \lim_{n \rightarrow \infty} \frac{9}{n^2} \frac{n\theta(3-\theta)}{9} = \lim_{n \rightarrow \infty} \frac{\theta(3-\theta)}{n} = 0$$

Then W is convergent.

4) Compare the two estimators T and W . Which one would you choose? Explain your choice.

We remark that $\frac{\text{Var}(W)}{\text{Var}(T)} - 1 = \frac{\frac{\theta(3-\theta)}{n}}{\frac{1+6\theta-3\theta^2}{3n}} - 1$

(95)

$$= \frac{3\theta(3-\theta) - (1+6\theta-3\theta^2)}{1+6\theta-3\theta^2} = \frac{3\theta - 1}{-3\theta^2 + 6\theta + 1}$$

We study the sign of $\frac{3\theta - 1}{-3\theta^2 + 6\theta + 1}$

We have $-3\theta^2 + 6\theta + 1 = 0$; $\Delta' = 9 + 3 = 12$

$$\Rightarrow \theta_1 = \frac{-3 - \sqrt{12}}{-3} = \frac{3 + 2\sqrt{3}}{3} > 0 \text{ and } \theta_2 = \frac{3 - 2\sqrt{3}}{3} < 0$$

Then $\forall \theta \in [0, 1]$, $-3\theta^2 + 6\theta + 1 > 0$ and the sign of $\frac{3\theta - 1}{-3\theta^2 + 6\theta + 1}$ is the sign of $3\theta - 1$ on $[0, 1]$ (1)

Thus

	θ	0	$\frac{1}{3}$	1
$\frac{\text{Var}(W)}{\text{Var}(T)} - 1$		-	0	+

Then if $\theta \in [0, \frac{1}{3}[$; $\frac{\text{Var}(W)}{\text{Var}(T)} - 1 < 0$ (95')

$\Rightarrow \text{Var}(W) < \text{Var}(T)$ then W is more efficient than T.

if $\theta \in]\frac{1}{3}, 1]$; $\frac{\text{Var}(W)}{\text{Var}(T)} - 1 > 0$ (95)

$\Rightarrow \text{Var}(W) > \text{Var}(T)$ then T is more efficient than W.